Cryptocurrency Investment Forecasting

Project - 1 (MA47201) report submitted to Indian Institute of Technology Kharagpur in partial fulfillment for the award of the degree of Bachelor of Science in Mathematics and Computing

by

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Under the supervision of

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Autumn Semester, 2024-25

November 15, 2024

DECLARATION

I certify that

A. The work contained in this report has been done by me under the guidance of my supervisor.

B. The work has not been submitted to any other Institute for any degree or diploma.

C. I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

D. Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Date: November 15, 2024 (Aryan Garg)

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CERTIFICATE

This is to certify that the project report entitled “Cryptocurrency Investment Forecasting” submitted by Aryan Garg (Roll No. 21MA10013) to Indian Institute of Technology Kharagpur towards partial fulfillment of requirements for the award of degree of Bachelor of Science in Mathematics and Computing is a record of bona fide work carried out by him under my supervision and guidance during Autumn Semester, 2024-25.

Date: November 15, 2024
Place: Kharagpur

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ABSTRACT

Name of the student: **Aryan Garg (21MA10013)**

Degree for which submitted: **Bachelor of Science - Mathematics and Computing**

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Thesis supervisor: **Prof. Somesh Kumar**

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This project focuses on leveraging time-series forecasting techniques to provide actionable investment insights in the cryptocurrency domain, particularly for Bitcoin and Ethereum. With access to historical price data, the project employs models such as ARIMA, Rolling Forecast ARIMA, XGBoost, and LSTM to predict future cryptocurrency prices. The LSTM model, a deep learning approach well-suited for volatile and non-stationary data, emerged as the most effective, delivering accurate predictions using a 60-day input window. The analysis highlights that Ethereum offers higher potential returns compared to Bitcoin after analyzing the logarithmic returns, providing valuable guidance for investment decisions. By capturing trends and short-term fluctuations in cryptocurrency markets, this study demonstrates the applicability of advanced forecasting techniques in volatile financial environments and sets the foundation for future work, including the exploration of additional cryptocurrencies and hybrid modeling approaches.

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1 Introduction

1.1 The Growing Relevance of Cryptocurrency Investments

Cryptocurrency has rapidly evolved into a significant asset class, drawing widespread attention from investors and financial institutions alike. Known for its unique qualities, including decentralization, transparency, and potential for high returns, the cryptocurrency market is reshaping the landscape of global finance. Unlike traditional assets, cryptocurrencies operate in a highly volatile and unpredictable environment, which presents both opportunities and challenges for investors. As digital currencies like Bitcoin and Ethereum continue to gain traction, the ability to forecast their price trends has become increasingly valuable.

1.2 The Need for Accurate Price Prediction

In a market characterized by frequent and dramatic price swings, accurate price prediction is essential for investors seeking to make data-driven decisions. With well-timed predictions, investors can better navigate the ups and downs of the cryptocurrency market, identify favorable entry and exit points, and manage risks effectively. Reliable forecasting can provide investors with a strategic advantage, enabling them to capitalize on short-term fluctuations and optimize their long-term portfolio strategies. Consequently, developing robust prediction models for cryptocurrency prices has become a key focus area in financial research.

1.3 Leveraging Data-Driven Models in Forecasting

The project utilizes historical data and statistical modeling to predict price movements in leading cryptocurrencies, namely Bitcoin and Ethereum. By employing advanced machine

learning and time-series forecasting techniques, this project aims to analyze and capture patterns within cryptocurrency price data. Models like ARIMA, rolling forecast ARIMA, Long Short-Term Memory (LSTM) networks, and XGBoost are explored for their potential to provide accurate and timely predictions. Each model offers distinct capabilities, and the project leverages these strengths to evaluate their effectiveness in handling the complex, volatile nature of cryptocurrency data.

1.4 Advantages of a Comprehensive Forecasting Approach

Predicting cryptocurrency prices presents unique challenges due to factors like market volatility, rapid shifts in sentiment, and susceptibility to external events. A data-driven approach allows for a systematic analysis of historical trends, extracting insights that can guide investment decisions. By combining different forecasting models and leveraging deep learning techniques, the project offers a multi-faceted approach to prediction, which enhances its robustness and reliability. This comprehensive forecasting system equips investors with the tools they need to make well-informed decisions, helping them manage risks and uncover potential opportunities within the cryptocurrency market.

1.5 Aiming to Empower Investors in a Volatile Market

The ultimate goal of this project is to empower investors by providing a reliable framework for cryptocurrency investment forecasting. With enhanced predictability, investors can navigate the challenges of a volatile market more effectively. By using a structured, data-driven methodology, this project provides insights that are critical for investment strategies in the ever-evolving cryptocurrency landscape. As the digital asset market continues to mature, such forecasting tools offer a valuable edge for investors seeking to harness the potential of cryptocurrencies while managing associated risks.

2 Literature Review

2.1 Time Series Forecasting in Financial Markets

Time series forecasting has been extensively researched for financial market applications, given the data's sequential nature and dependence on past observations. Techniques such as ARIMA (Autoregressive Integrated Moving Average) have historically shown promise in forecasting market trends and prices. Machine learning methods such as Long Short-Term Memory (LSTM) networks excel at learning complex temporal dependencies and long-range patterns, making them well-suited for non-linear and highly volatile datasets. Similarly, gradient boosting models like XGBoost are valued for their ability to incorporate engineered features and adapt to structured data efficiently. Extensions to classical forecasting models (like ARIMA), such as rolling forecast strategies, enable continuous updates to predictions by recalibrating on new data, enhancing their adaptability in dynamic market conditions. These methodologies offer diverse approaches for addressing the challenges posed by financial time series data, particularly in cryptocurrency markets where price movements are influenced by a combination of historical trends and external factors.

2.2 The ARIMA Model and Its Application

The ARIMA model, traditionally used in forecasting economic indicators, has been applied to cryptocurrency markets. ARIMA is particularly useful when time series data exhibit seasonality and trends. It is structured to account for past values and shocks to the system, making it suitable for capturing some of the regular patterns seen in financial markets. However, cryptocurrency data often displays extreme volatility, and ARIMA's limitations in modeling highly non-linear relationships reduce its effectiveness in this context.

2.3 Rolling Forecast ARIMA Model

The Rolling Forecast ARIMA model is an extension of the traditional ARIMA model, designed to enhance its forecasting capabilities by adapting to dynamic data patterns.. A Rolling Forecast ARIMA model is a time series forecasting approach where the model makes predictions one step (or day) at a time. Unlike static ARIMA models, which rely on a fixed training dataset, the rolling forecast strategy ensures that the model continuously incorporates the latest information, making it particularly suitable for highly volatile and non-stationary time series, such as financial markets. However, since this model predicts rolling day by day increments, it isn't fair to compare it to the other models.

2.4 Introduction of LSTM Models in Financial Forecasting

With the limitations of traditional models in mind, researchers have shifted towards machine learning techniques. One prominent model is Long Short-Term Memory (LSTM), a type of Recurrent Neural Network (RNN) designed to handle long-term dependencies in sequential data. LSTM models excel in scenarios where data sequences contain significant temporal dependencies, which is common in cryptocurrency price movements. Architecture of LSTM is suitable for predicting cryptocurrency prices, as it can capture both short-term volatility and long-term trends, handling the chaotic nature of financial time series more effectively than traditional models. LSTMs employ memory cells and gating mechanisms to retain relevant information while discarding less useful data.

Cryptocurrency price forecasting benefits significantly from LSTMs, given their ability to model complex, non-linear relationships. Unlike traditional models like ARIMA, LSTMs excel in capturing patterns influenced by multiple factors, making it a robust choice for cryptocurrency prediction tasks.

2.5 XGBoost for Time Series Prediction

XGBoost (Extreme Gradient Boosting) is a machine learning algorithm known for its scalability and efficiency. It constructs an ensemble of decision trees through gradient boosting, optimizing performance and reducing overfitting. While not inherently designed for time series data, XGBoost can be adapted for forecasting by incorporating lagged variables and temporal features like day-of-week or month-of-year. XGBoost, known for its speed and accuracy, is particularly effective with tabular datasets and has been utilized to predict financial time series by analyzing a range of features, including historical prices, volume, and technical indicators.

For cryptocurrency forecasting, XGBoost offers the advantage of incorporating multiple explanatory variables beyond past price data. Studies have demonstrated its utility in capturing interactions among features, making it an effective tool for predicting market movements.

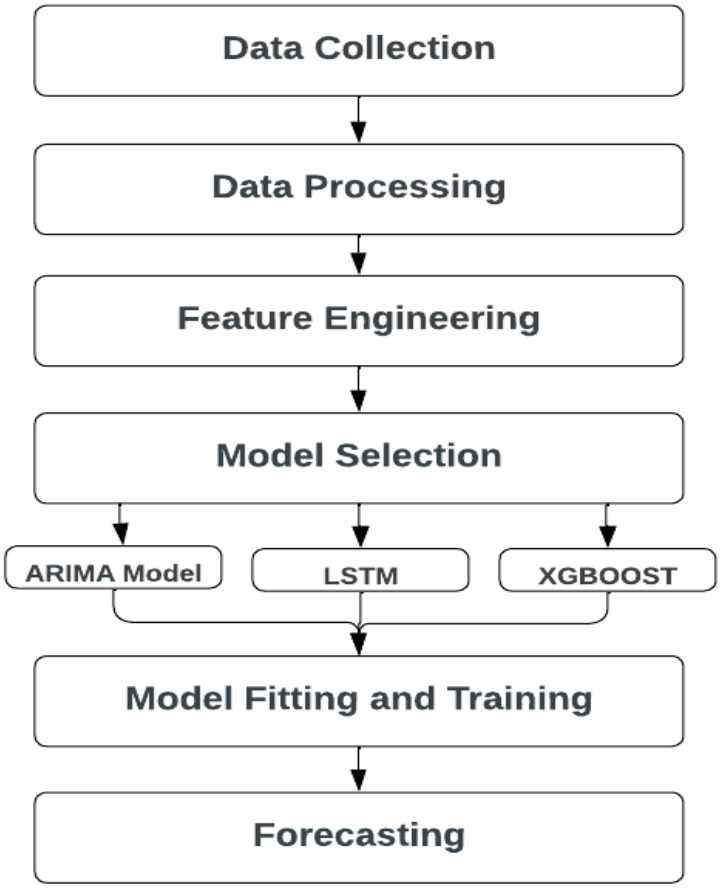
2.6 Comparative Insights

While ARIMA provides a solid statistical foundation for forecasting stationary time series, its reliance on linear relationships may limit its effectiveness in modeling complex dynamics. Rolling Forecast ARIMA addresses this by updating the training dataset continuously, improving its adaptability to changing data.

LSTMs, on the other hand, excel in capturing non-linear dependencies and temporal patterns, making them suitable for highly volatile and non-stationary data like cryptocurrency prices. XGBoost complements these models by leveraging its strength in feature engineering and handling non-sequential explanatory variables.

3 Methodology

Accurate forecasting of cryptocurrency prices is essential for making informed investment decisions and managing financial risk. One effective approach to time series forecasting is through advanced machine learning and statistical models such as ARIMA, Rolling Forecast ARIMA, LSTM, and XGBoost. In this section, we discuss the methodology for predicting Bitcoin and Ethereum prices by leveraging these models, highlighting the data preparation, feature engineering, model selection, and evaluation processes.



3.1 Data Collection and Preprocessing

The dataset used for the project is sourced from Yahoo Finance and contains daily historical price data for Bitcoin and Ethereum. The Dataset for Bitcoin is over a span of 14 years and for Ethereum is over a span of around 7 years. The data includes features such as Open, High, Low, Close, Volume, and Adjusted Close prices. Dataset was properly preprocessed and handled the dataset rows which were null in any of the fields present.

Combining the four price types (Open, High, Low, Close) into a single Weighted Price is a practical approach to simplify the forecasting target. A common formula for weighted price is:

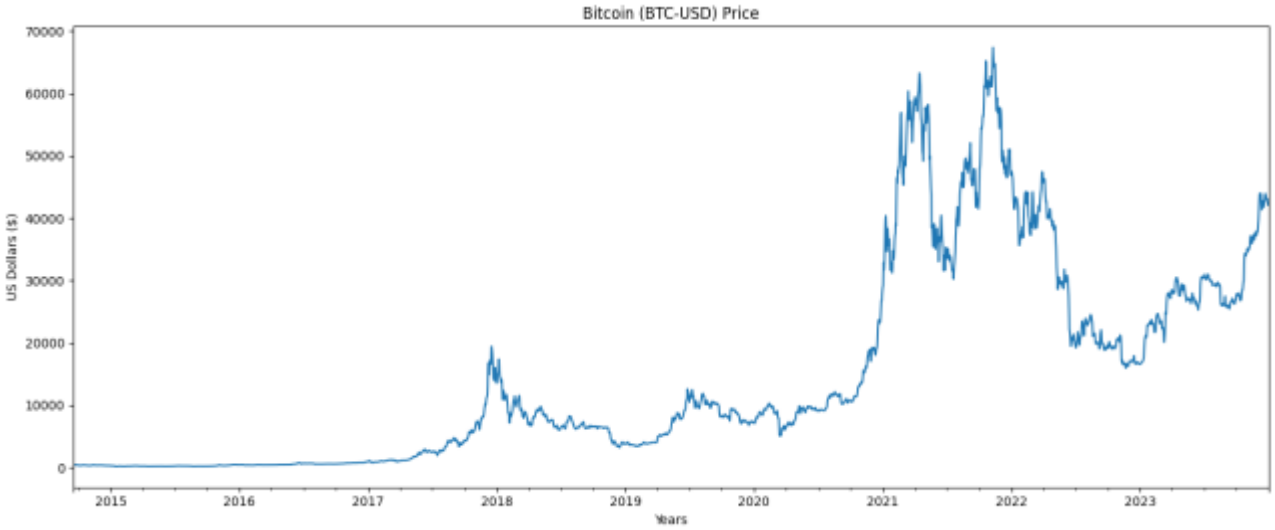
$$Weighted\ Price\ =\ (High\ +\ Low\ +\ Open\ +\ Close)/4$$

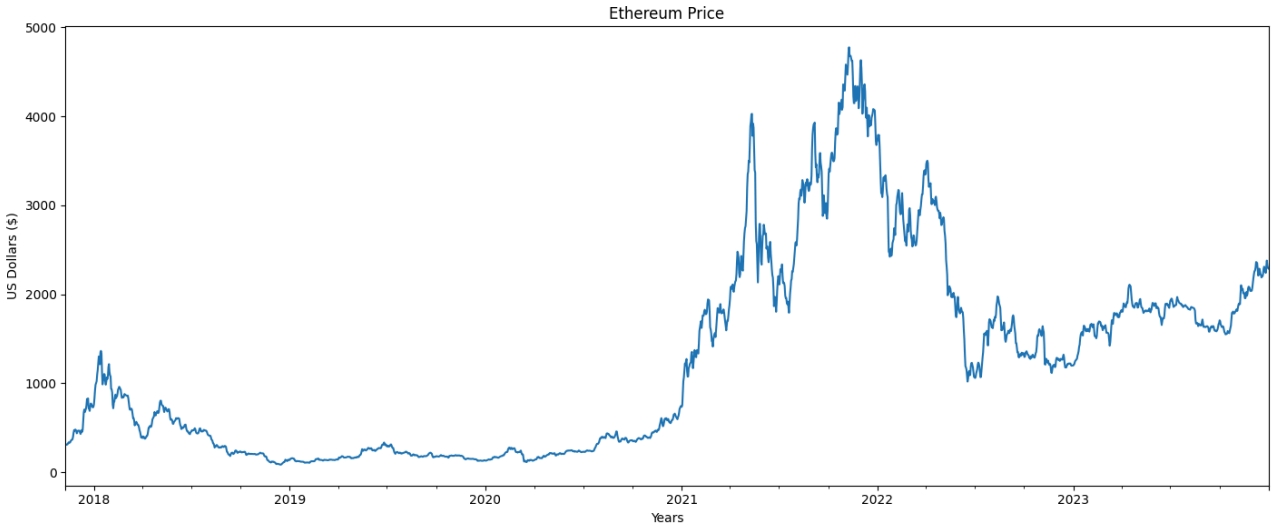
Choosing equal weights for calculating the weighted price ensures simplicity, neutrality, and a balanced representation of all aspects of daily market activity (Open, High, Low, Close). In some models, reshaping and scaling the dataset is also required to meet the input for the model as in LSTM. LSTM models tend to perform better when the data is scaled, typically between 0 and 1

3.2 Feature Engineering

Feature Engineering refers to the process of creating, modifying, or selecting features (input variables) and also selecting the precise amount of dataset that enhance the performance of a machine learning model. Features are the measurable properties or characteristics of the phenomenon being observed, and the quality of these features directly affects the model's ability to make accurate predictions.

Features like "Day of the Week", "Month", "Year", etc. needed to be extracted from the “Date” field while using the XGBoost Model and generating new features from the raw data make the data more informative. Data set needed to be constricted for both Bitcoin and Ethereum as that did not show much volatility and would hinder the model’s ability to make accurate predictions.





3.2 Model Selection

Three forecasting models were selected for implementation: ARIMA, Rolling Forecast ARIMA, and LSTM, with XGBoost as a supplementary machine learning model.

ARIMA: Chosen for its ability to handle linear trends and seasonality in time series data. The model parameters (p, d, q) were determined using the ACF and PACF plots.

Rolling Forecast ARIMA: An extension of ARIMA that updates the training data dynamically, allowing predictions to adapt to recent trends. While using the ARIMA model, it came to a surprise that training data can be updated and this was used to predict using this model and the parameters were kept the same as before.

LSTM: A neural network model tailored for sequential data. It captures long-term dependencies and temporal relationships effectively. The LSTM model was built on 2 hidden layers for proper functioning along with dense layers. It would increase the model's training time but would help in accurate predictions.

XGBoost: To use the XGBoost model, some features needed to be introduced as mentioned in the feature engineering which enhances the model's working. The features here represent different time-based factors (e.g., the day of the week, month, and year). Features like DayOfWeek, DayOfMonth, etc., help XGBoost understand time-related patterns in the Bitcoin and Ethereum prices.

3.3 Model Fitting and Training

The dataset was split into training (70%) and testing (30%) subsets.

ARIMA and Rolling Forecast ARIMA: Models were trained iteratively on rolling windows of data. Parameters were chosen keeping in mind the non stationary leading to differencing as 1, ACF plot showed no such patterns leading to value of q as 0, PACF plot showed significant lags up to 5.

LSTM: A sliding window approach was used to create sequences of input-output pairs. Hyperparameters, such as the number of neurons, layers, and learning rate, were tuned to optimize performance. The dataset needed to be reshaped as stated before and use of 3 layers leads to increase in time but the model is properly trained. LSTM layers use return_sequences=True to retain temporal information for subsequent layers, while the final LSTM layer outputs only the last hidden state (return_sequences=False). ReLU activation is used for its simplicity and faster convergence and its ability to deal with large datasets. It outputs the input directly if it's positive, and zero otherwise:

$$f(x) = max(0, x).$$

XGBoost: XGBRegressor is used with the objective of minimizing the squared error (reg:squarederror) and keeping the value of estimators high (1000) to improve the performance and using a lower learning rate prevents the model from overfit. These parameters allow fine tuning of the model.

3.4 Forecasting

The model we used has been trained and fitted using the inbuilt python libraries for the respective model. The respectives trained models are used to predict the values on the test data for Bitcoin and Ethereum. LSTM and XGBoost required preprocessing of the test data into the type of input data for predictions. The predicted data again needed to be rescaled and reshaped to get the data to be comparable to the actual test data to do observations and make conclusions regarding the best fit model for predicting cryptocurrencies.

The forecasts were plotted alongside actual values to visualize the performance and identify deviations. Root Mean Squared error is also calculated after predictions with respect to the actual data and that would help us derive conclusions.

3.5 Best Fit model

Depending on the RMSE, best fit models are determined and the best model is used to make future predictions and draw the comparisons between Bitcoin and Ethereum and logarithmic returns are calculated to determine which cryptocurrency would be the best to invest in and would give us higher returns.

4 Results and Observations

Root Mean Squared Error (RMSE) was used as an evaluation criteria for finding out the best model suited for predictions. RMSE was calculated for all the 4 models used.

$$RMSE = sqrt\,[(\Sigma(Pi - Oi)^2) / n]$$

4.1 Bitcoin

The **Long Short-Term Memory (LSTM) model** comes out to be the best fit model for predicting future prices of Bitcoin with an RMSE of around 900.

The RMSE for the four models used are as follows:

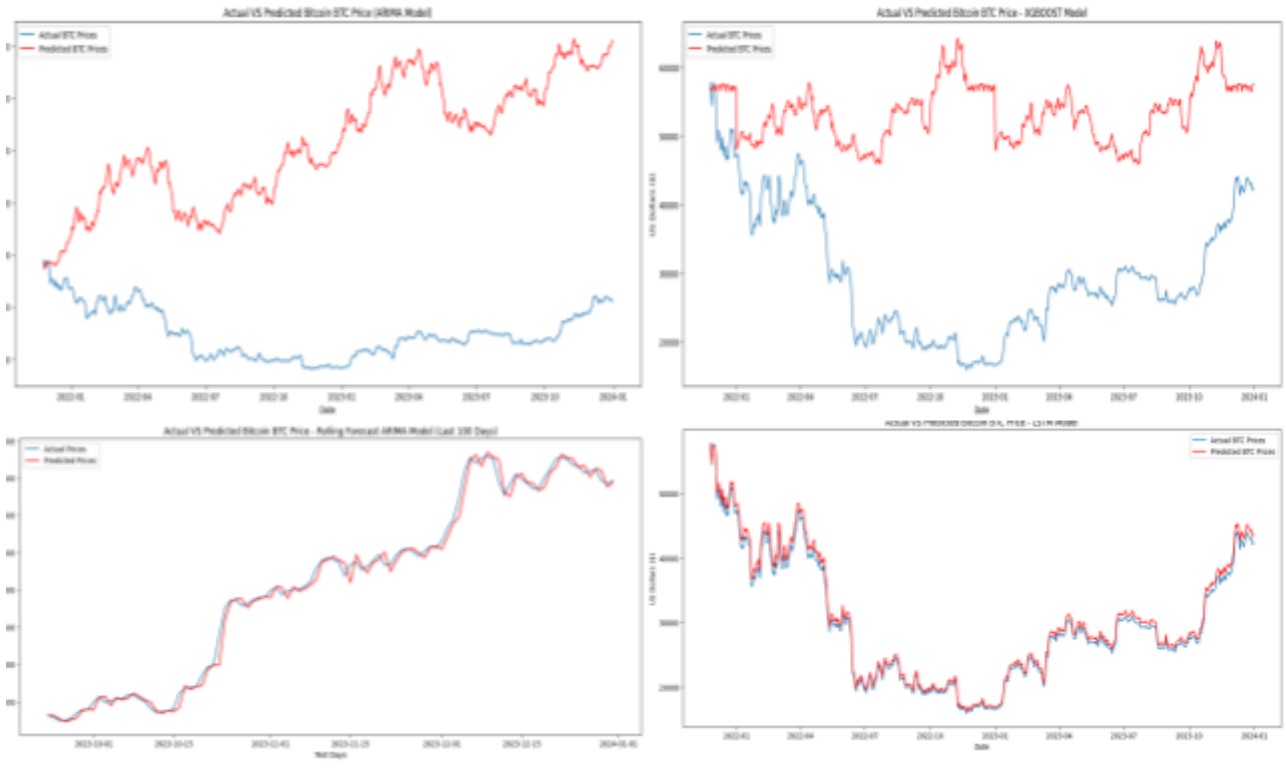
$$ARIMA: 77628.313$$

$$Rolling\ Forecast\ ARIMA: 589.074$$

$$LSTM: 829.669$$

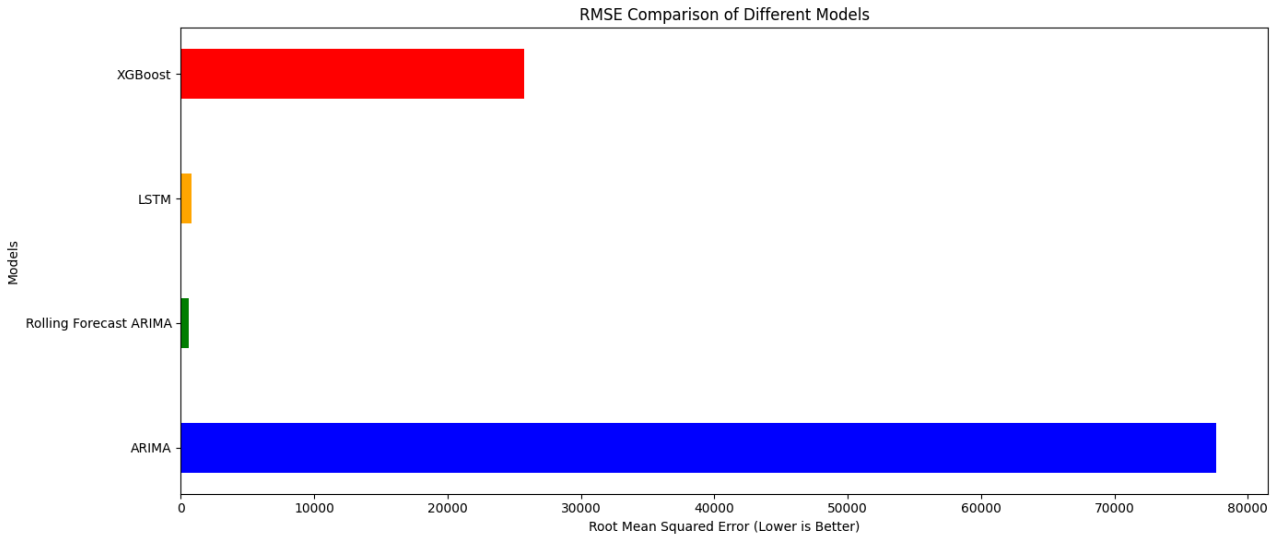
$$XGBoost: 25765.605$$

The 4 models predicting prices were plotted alongside actual test data and following is the plot:



- Rolling Forecast ARIMA model is not fair to compare to other models as the model predicts in rolling day by day increments and due to its sequential prediction nature. Hence, LSTM model has the lowest RMSE and best predictions
- ARIMA model does not give accurate results as it does not go very well with high volatile and non stationary data such as the dataset for bitcoin.
- LSTM models give very accurate results as it uses neural networks and it makes them a deep learning network well suited for time series forecasting. This is attributed to its ability to capture complex temporal patterns and dependencies using neural networks. The use of two hidden layers and a dense output layer further improved the model's accuracy, making it well-suited for time series forecasting of high-variance data.
- The XGBoost model, while typically a strong performer for structured datasets, did not yield the expected accuracy in this case. This can be attributed to the dataset's unstructured nature and the inherent challenges of predicting highly volatile cryptocurrencies.

Following is the plot of RMSE comparisons for the four models used for predictions:



4.2 Ethereum

The **Long Short-Term Memory (LSTM) model** comes out to be the best fit model for predicting future prices of Ethereum with an RMSE of around 39.

The RMSE for the four models used are as follows:

$$ARIMA: 2125.826$$

$$Rolling\ Forecast\ ARIMA: 27.804$$

$$LSTM: 39.904$$

$$XGBoost: 454.262$$

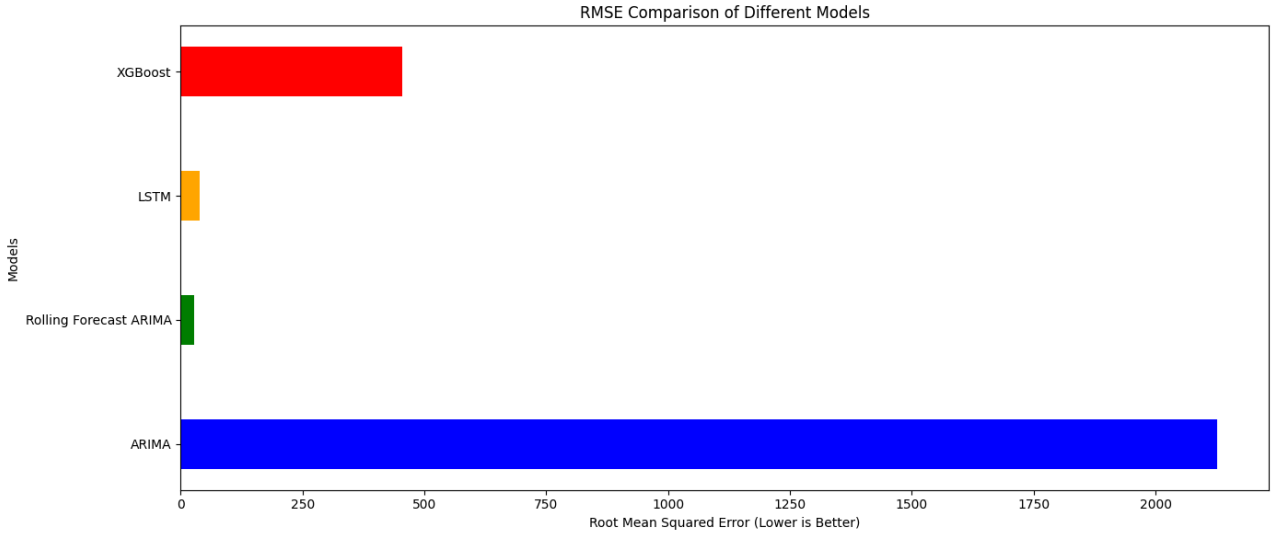
The 4 models predicting prices were plotted alongside actual test data and following is the plot:



- Rolling Forecast ARIMA model is not fair to compare to other models as the model predicts in rolling day by day increments and due to its sequential prediction nature. Hence, LSTM model has the lowest RMSE and best predictions
- ARIMA model does not give accurate results as it does not go very well with high volatile and non stationary data such as the dataset for Ethereum.
- LSTM Model's ability to capture complex temporal patterns and dependencies using neural networks gave very fine results, making it well-suited for time series forecasting of high-variance data.

- XGBoost did not predict well as in the same case as predictions for bitcoin were made due to the not so well structured dataset of ethereum.

Following is the plot of RMSE comparisons for the four models used for predictions:



4.3 Bitcoin v/s Ethereum

The LSTM model, identified as the best fit for both Bitcoin and Ethereum, was utilized to make future predictions. The logarithmic returns for Bitcoin and Ethereum were plotted simultaneously, revealing distinct trends. A 60-day window was selected for predictions, as it balances the trade-off between capturing relevant recent trends, seasonality, and short-term fluctuations, while avoiding the challenges of overfitting or overwhelming the model with excessive historical data. This choice also optimizes computational efficiency, given the high resource demands of LSTM models with longer input sequences.

From the plotted results, it becomes evident that Ethereum demonstrates a potential for higher returns compared to Bitcoin for the same amount of investment. This conclusion highlights Ethereum's comparatively volatile yet profitable nature within the observed time period.

Following is the plot of weighted price of future Bitcoin and Ethereum as well as the logarithmic returns of the same for the first 15 days:



Ethereum has a logarithmic return of up to 0.6 in the initial 15 days.

5 Conclusion and Future Work

5.1 Conclusion

This project aimed to forecast cryptocurrency prices, focusing on Bitcoin and Ethereum,

using advanced time series models such as ARIMA, Rolling Forecast ARIMA, XGBoost, and LSTM. Among these, the LSTM model emerged as the best fit due to its ability to effectively capture complex patterns in highly volatile and non-stationary data. The LSTM model's predictions exhibited the lowest RMSE, highlighting its superiority over traditional models like ARIMA and machine learning models like XGBoost. This is primarily due to its deep learning architecture, which is well-suited for sequential data.

The logarithmic returns for both Bitcoin and Ethereum were analyzed. Based on the results, Ethereum is identified as the more profitable investment choice, offering higher returns compared to Bitcoin over the observed period.

5.2 Future Work

The scope of this project can be expanded in several directions to improve and diversify the forecasting approach:

- *Inclusion of Additional Cryptocurrencies:* Expanding the analysis to include emerging and rapidly growing cryptocurrencies, such as Solana, Cardano, and Binance Coin, to capture broader market trends and identify new investment opportunities.
- *Exploring Hybrid Models:* Combining statistical, machine learning, and deep learning methods.
- *Feature Engineering Enhancements:* Introducing additional features like trading volume, social media sentiment, macroeconomic indicators.
- *Cross-Asset Analysis:* Studying the relationship between cryptocurrencies and traditional assets such as gold, stocks, and bonds to explore potential diversification strategies.

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